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Periodic Defect Engineering of Iron–Nitrogen–Carbon Catalysts for Nitrate Electroreduction to Ammonia

*Runxi Zhu, Yanyang Qin, Tiantian Wu, Shujiang Ding, and Yaqiong Su**

Iron–nitrogen–carbon single atom catalyst (SAC) is regarded as one of the promising electrocatalysts for NO_3^- reduction reaction (NO_3RR) to NH_3 due to its high activity and selectivity. However, synergistic effects of topological defects and FeN_4 active moiety in Fe–N–C SAC have rarely been investigated. By performing density functional theory (DFT) calculations, 13 defective graphene FeN_4 with 585, 484, and 5775 topological line defects are constructed, yielding 585-68-FeN_4 with optimal NO_3RR catalytic activity, high selectivity, as well as robust anti-dissolution stability. The high NO_3RR activity on 585-68-FeN_4 is well explained by the high valence state of Fe center as well as asymmetric charge distribution on FeN_4 moiety influenced by 5- and 8-member rings. This DFT work provides theoretical guidance for engineering NO_3RR performance of iron–nitrogen–carbon catalysts by modulating periodic topological defects.

1. Introduction

Ammonia (NH_3), as a source of nitrogen for fertilizer and a potential carbon-free energy carrier, is an indispensable chemical for modern industrial development.[1] The industrial-scale NH_3 synthesis is still dominated by Haber–Bosch process operating under harsh conditions with nitrogen and methane-reformed hydrogen as feedstocks, which however is not only an energy- and capital-intensive approach but also suffers from extremely low NH_3 yields.[2] In recent years, ambient electrocatalytic N_2 reduction reaction has emerged as an appealing alternative, which however suffers from unsatisfactory Faradaic efficiency and low NH_3 yields due to extremely stable triple N≡N bond (941 kJ mol^{-1}) and low N_2 solubility.[3] Therefore, using N_2 as the N source for electrochemical NH_3 synthesis to achieve considerable yields and wider application remains challenging.

Nitrate (NO_3^-), alternative to the inert N_2, is considered as an attractive N source for electrochemical synthesis of NH_3 due to the much weaker N═O bond (204 kJ mol^{-1}) and N—O bond (176 kJ mol^{-1}) than N≡N bond (941 kJ mol^{-1}).[4] Notably, NO_3^- is one of the most widespread water pollutants due to the discharge of industrial wastewater and chemical fertilizers, posing potential threats to human health.[5] Therefore, electrocatalytic NO_3^- reduction reaction is regarded as a promising method of removing NO_3^- contaminants from wastewater. Furthermore, NO_3^--reduction-reaction (NO_3RR)-based cathode and metal anode can be assembled as a metal–NO_3^- battery for electricity production. Recently, Zhang et al. reported that the iron-doped nickel phosphide (Fe/Ni_2P) catalyst exhibits high NH_3 Faradaic efficiency (FE) and NO_3^- conversion efficiency.[6] And a Zn–nitrate battery is further assembled with a power density of 3.25 mW cm^{-2} and a FE of 85.0% for NH_3 production. Evidently, the NO_3^--to-NH_3 conversion can not only deliver both electricity and NH_3 products, but also remove NO_3^- contaminants from wastewater.

The NO_3^--to-NH_3 conversion involves nine proton and eight electron transfers and is accompanied by many possible reaction pathways and byproducts (NO_2, NO, N_2O, N_2, NH_2OH, etc.). As a predominant competition, NO_3^- reduction to N_2 pathway involves a N—N coupling step, where two neighboring active sites are possibly needed such as Rh- or Cu-based metal catalysts.[7] In order to improve the selectivity toward NH_3, dispersing transition metals into isolated single atoms can effectively suppress N—N coupling. For example, Wu et al. reported excellent activity and selectivity of Fe single atomic sites deposited on a standard glassy carbon electrode for NO_3^- reduction to NH_3 with a maximal ammonia Faradaic efficiency of ≈75% and a yield rate of up to ≈20 000 μg h^{-1} mg$_{cat.}^{-1}$.[8] In addition, FeN_4 is identified as the active site for NO_3RR on atomically dispersed Fe–N–C catalyst via density functional theory calculations. Nevertheless, the essential synthetic pyrolysis process of Fe–N–C catalysts may induce undesirable structures such as numerous topological defects, metal–N_x (x = 1–4) and metal–C in the carbon matrix.

It is noteworthy that the fabrication of a perfect single-layer graphene with absolutely 100% hexagonal lattice is very challenging because graphene produced by any method contains some specific defects or defects in graphene may be inherent.

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In addition, it is feasible to introduce specific defects artificially in controlled manner (irradiation or chemical treatments) to achieve the desired properties.[9] In summary, various types of defects in graphene make it possible to tailor the local properties of graphene, such as mechanical, electronic, and magnetic properties.[10] Some defects in graphene are well-known and have been experimentally observed via transmission electron microscopy (TEM), such as Stone–Wales (SW) defect or 5-8-5 divacancy defect.[11] A SW defect (also known as 5775 defect) is a crystallographic defect which is formed by an in-plane 90° rotation of the neighboring C—C pairs relative to the midpoint of their bond.[10b] In this way, after the rotation of C—C pairs in perfect graphene, four hexagons are converted into two 5-member rings and two 7-member rings. A 5-8-5 divacancy defect is formed by the removal of the neighboring C—C pairs, after which double vacancy is formed and reconstructed, resulting in two 5-member rings and one 8-member ring labeled the 5-8-5 defect.[12] Likewise, A 4-8-4 defect is formed by the continuous removal of the multiple neighboring C—C pairs.[13]

The defect engineering is recognized as an effective method to promote the electrocatalytic performance of carbon materials.[14] For example, Jia et al. reported that defects derived by the removal of N-doped atoms from graphene are demonstrated, both experimentally and theoretically, to be effective for oxygen reduction reaction, oxygen evolution reaction (OER), and hydrogen evolution reaction (HER).[15] Various structural defects (5-,7-, and 8-member rings) with different combinations such as 585, 75585, and 5775 were observed via TEM. They confirmed that the topological defects at the edge as active sites can significantly enhance the catalytic activities. In addition, Zhang et al. demonstrated both experimentally and theoretically that the graphene defects (5775 defect or divacancy) trap atomic Ni species (aNi) inside to form an integrity (aNi@defect) as the active site toward HER and OER, which shows the synergistic effects of graphene defects and aNi for boosting the HER and OER performances.[16] As mentioned above, atomically dispersed Fe–N–C single atom catalyst exhibits excellent NO_3RR performance. The synthetic pyrolysis process of Fe–N–C may induce numerous defects on the graphene or around the FeN_4 site. However, either FeN_4 site or defect is explored separately, and the synergetic effects between them is rarely investigated.

In this work, we provide theoretical insights into the effects of periodically topological defects on FeN_4 active site for NO_3RR by means of density functional theory calculations. First, three types of defective graphene with 585, 484, and 5775 topological line defects are constructed as the substrate model, and FeN_4 moiety is anchored on topological line defect forming 13 kinds of defective graphene FeN_4 (DG-FeN_4) moieties. Also, NO_3RR performance on pristine graphene FeN_4 (pristine FeN_4) is investigated, where the first hydrogenation step is identified as the potential-determining step (PDS) with a $\Delta G_{\max}$ of 0.41 eV. Among 13 kinds of DG-FeN_4, 8 kinds of DG-FeN_4 are able to reduce the energy barrier of PDS, exhibiting more excellent NO_3RR electrocatalytic activity than pristine FeN_4. Notably, 585-68-FeN_4 presents the optimal NO_3RR catalytic performance, with a $\Delta G_{\max}$ value as low as 0.22 eV. Furthermore, we evaluate the competition between NO_3RR and HER on 8 kinds of DG-FeN_4 via the comparison of $\Delta G(^*NO_3)$ versus $\Delta G(^*H)$ and $\Delta G_{\max}$ of NO_3RR versus $\Delta G(^*H)$, yielding 585-68-FeN_4 with high selectivity as well as activity. Anti-dissolution stability diagrams as a function of pH and potential on pristine FeN_4 and 585-68-FeN_4 at different Fe^{2+} concentrations are plotted, showing that 585-68-FeN_4 presents a broader anti-dissolution region than pristine FeN_4. Differential charge density and Bader charge analysis calculations were conducted to investigate the electron transfer and redistribution between Fe and N affected by different topological defects. It is proved that the high NO_3RR activity on 585-68-FeN_4 is ascribed to high valence state of Fe and asymmetric charge distribution on FeN_4 moiety.

2. Results and Discussion

2.1. Structure and Stability

As shown in **Figure 1**a–c, 3 types of defective graphene with 585, 484, and 5775 topological line defects were constructed as the substrate model. Furthermore, we calculated the density of states (DOS) of pristine graphene, DG-585, DG-484, and DG-5775, as shown in Figure S1 (Supporting Information). It can be seen that DG-585, DG-484, and DG-5775 possess increased DOS values than pristine graphene near the Fermi level, corresponding to their enhanced metallic conductivity.[17] Figure S2 (Supporting Information) plots the partial electron densities (PEDs) near the Fermi level for pristine graphene, DG-585, DG-484, and DG-5775, respectively. Apparently, each carbon atom in 585, 484, and 5775 topological line defects shows a much larger charge distribution perpendicular to the plane. To anchor Fe single atom, the 2 adjacent carbon atoms were removed and the 4 nearest-neighbored carbon atoms were replaced with N atoms. Considering the shape and position of the carbon ring, it is observed in Figure 1a that there are 4 ways to remove 2 adjacent carbon atoms in 585-defective graphene, denoted by 585-55, 585-56, 585-58, and 585-68, respectively. Similarly, Figure 1b shows that there are 4 ways to remove 2 adjacent carbon atoms in 484-defective graphene, denoted by 484-46, 484-48, 484-68-1, and 484-68-2, respectively. As shown in Figure 1c, there are 5 ways to remove 2 adjacent carbon atoms in 5775-defective graphene, denoted by 5775-56, 5775-57-1, 5775-57-2, 5775-67, and 5775-77, respectively. As a result, 13 kinds of DG-FeN_4 moieties were obtained. Figure 1d displays the optimized structures of 585-55-FeN_4, 484-46-FeN_4, and 5775-56-FeN_4. The optimized structures of pristine FeN_4 and the remaining 10 kinds of DG-FeN_4 are shown in Figures S3–S5 (Supporting Information).

To evaluate the stabilities of Fe single atom catalysts (SACs) in different graphene substrates, the binding energies (E_b) between Fe single atom and the adjacent 4 N atoms were calculated, as shown in Figure 1e. E_b is below zero in the range from −0.99 to −4.09 eV (Table S1, Supporting Information), indicating that Fe single atoms can be stably embedded in different graphene substrates and form strong coordination bonds with the adjacent 4 N atoms. In addition, we also examine the formation energies of all structures, as shown in Figure 1e. It can be seen that only 5775-56 (2.12 eV) and 5775-57-1 (1.72 eV) possess higher formation energies than pristine FeN_4 (1.71 eV), indicating that the remaining 11 kinds of DG-FeN_4 can be synthesized easier than pristine FeN_4 in the experiment.

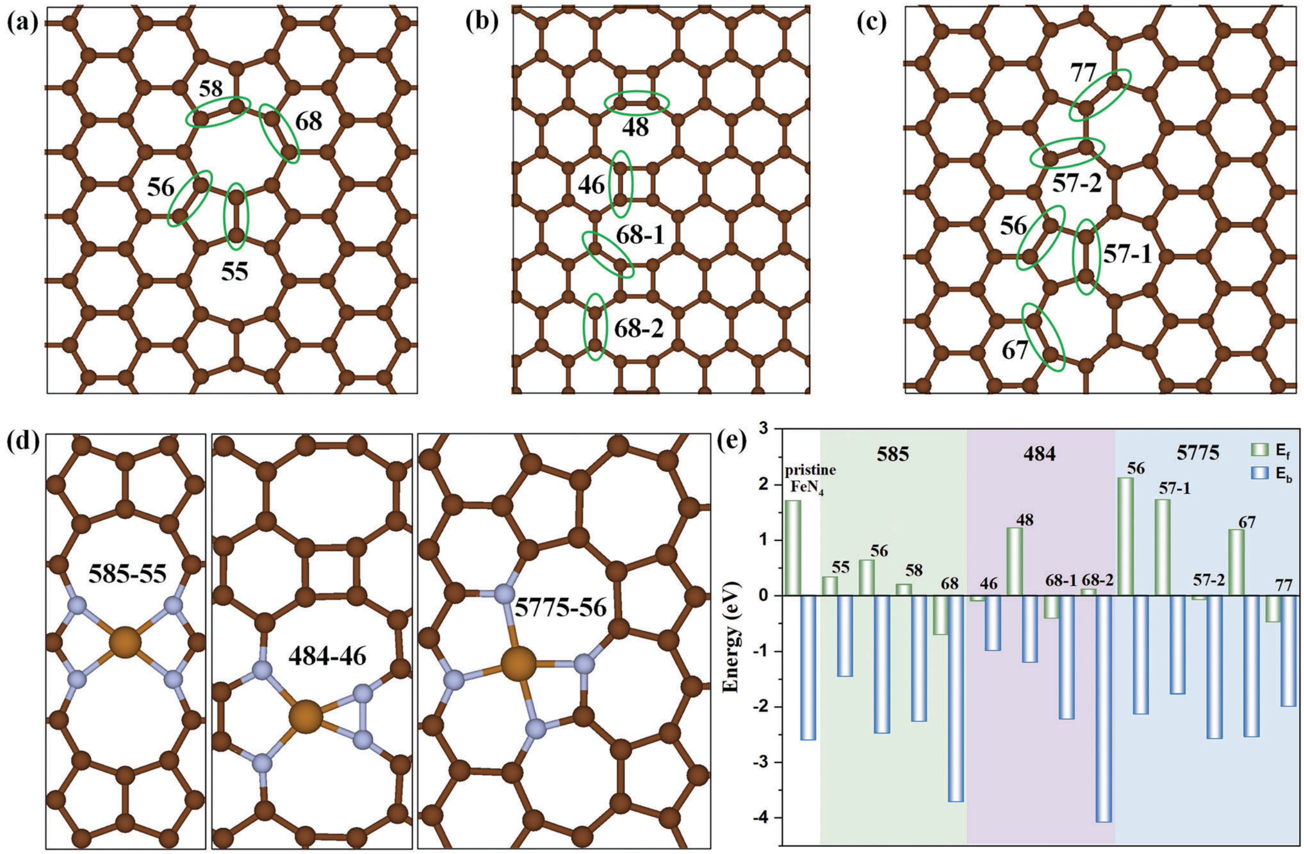

Figure 1. Structure and stability of pristine graphene FeN_4 and defective graphene FeN_4. The top views of optimized structure of a) 585-defective graphene, b) 484-defective graphene, c) 5775-defective graphene, and d) 585-55-FeN_4, 484-46-FeN_4, and 5775-56-FeN_4. Dark brown sphere: carbon; blue sphere: nitrogen; light brown sphere: iron. e) The binding energies (E_b) and formation energies (E_{form}) of pristine graphene FeN_4 and 13 kinds of defective graphene FeN_4.

2.2. NO_3RR Activity on Fe SACs

The conversion of NO_3^- to NH_3 ($NO_3^- + 9H^+ + 8e^- \rightarrow NH_3 + 3H_2O$) is accompanied by nine protons and eight electrons. After structure optimization, NO_3RR intermediates are all preferentially adsorbed on Fe single atom, as shown in **Figure 2**.

The initial step in NO_3RR process is the adsorption of NO_3^-, which plays a key role in the whole catalytic reaction. Figure S6 (Supporting Information) depicts two possible adsorption configurations of NO_3^- adsorbed on Fe single atom, namely 1-O and 2-O patterns. It is noted that 2-O pattern converts into 1-O pattern after structure optimization, indicating that 1-O

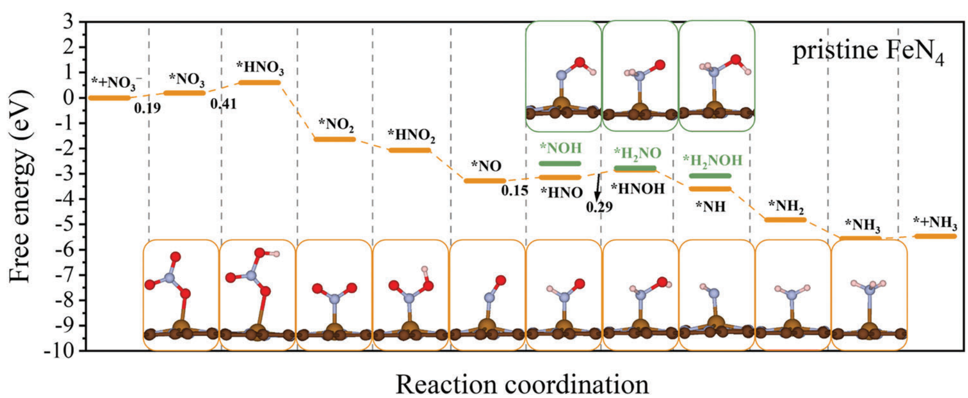

Figure 2. NO_3RR catalytic performance on pristine FeN_4. The Gibbs free energy diagrams for NO_3RR at 300 K when pH = 0, U = 0 V on the surfaces of pristine FeN_4. The insets denote the optimized structures of NO_3RR intermediates. Dark brown sphere: carbon; blue sphere: nitrogen; light brown sphere: iron; red sphere: oxygen; white sphere: hydrogen.

pattern is a more favorable adsorption configuration. To simplify the computation and find out the most suitable pathway on FeN_4, three possible adsorption configurations and the adsorption free energy of $^*NO_2$ and *NO are calculated and listed in Figures S7 and S8 (Supporting Information), namely N-end, O-end, and NO-side patterns. It is observed in Figure S7 (Supporting Information) that NO-side pattern of $^*NO_2$ converts into N-end pattern after structure optimization, and N-end pattern possesses a lower adsorption free energy of −1.68 eV than that of O-end pattern (−1.46 eV). Likewise, in Figure S8 (Supporting Information), NO-side pattern of *NO converts into N-end pattern after structure optimization, and N-end pattern possesses a lower adsorption free energy of −1.91 eV than that of O-end pattern (−0.56 eV). Therefore, N-end pattern is the most stable and favorable adsorption configuration for both $^*NO_2$ and *NO. In the subsequent reaction pathway, we will merely take N-end pattern into account. The Gibbs free energy diagrams of NO_3RR on pristine FeN_4 are displayed in Figure 2, where the reaction with the maximum ΔG (ΔG_{max}) is PDS. And the corresponding values of Gibbs free energy are listed in Table S2 (Supporting Information). The initial step is the adsorption of NO_3^-, with a ΔG of 0.19 eV. Subsequently, $^*NO_3$ is hydrogenated by ($H^+ + e^-$) pair to form $^*HNO_3$ intermediate, which is the potential-determining step for pristine FeN_4 with a ΔG_{max} of 0.41 eV. In the following hydrogenation steps, $^*HNO_3$ goes through a successive exothermal process to form $^*NO_2$, $^*HNO_2$, and *NO with ΔG of −2.24, −0.43, and −1.21 eV. When *NO is hydrogenated, it can form *HNO ($\Delta G = 0.15$ eV) or *NOH ($\Delta G = 0.69$ eV), demonstrating that *HNO is thermodynamically favorable. Therefore, we merely consider the hydrogenation of *HNO. Next, *HNO is attacked by H proton to form *HNOH with a ΔG of 0.29 eV, which is lower than that of $^*H_2NO$ ($\Delta G = 0.36$ eV). *HNOH is hydrogenated to form *NH or $^*H_2NOH$, and it is observed in Figure 2 that *NH is thermodynamically more favorable than $^*H_2NOH$. The yellow line in Figure 2 denotes the minimum energy pathway (MEP), in which *HNO, *HNOH, and *NH (*NO → *HNO → *HNOH → *NH) are thermodynamically more favorable than the other intermediates. In the subsequent steps (*NH → $^*NH_2$ → $^*NH_3$), H proton consecutively attacks intermediates to form $^*NH_2$ and $^*NH_3$. Besides, the desorption of $^*NO_2$ and *NO is very difficult, as indicated by the large desorption energy barriers of 1.79 and 1.45 eV, which are much higher than the energy barriers of the hydrogenation process. In other words, both $^*NO_2$ and *NO will be further hydrogenated rather than desorbed.

Based on MEP on pristine FeN_4 mentioned above, it is evident that the reactions of $^*NO_3$ → $^*HNO_3$, *NO → *HNO, and *HNO → *HNOH are all uphill steps. In other words, the ΔG values of these reactions are positive, indicating that they are thermodynamically endothermal processes. We refer to the above three reactions as key steps that will influence the PDS on FeN_4 active site. In order to elucidate the effects of topological defects on NO_3RR activity, we first examined the NO_3RR performance on DG-FeN_4 via three key steps, and the corresponding ΔG values of three key steps on 13 kinds of DG-FeN_4 are listed in Table S3 (Supporting Information). Our calculation results (Table S3, Supporting Information) indicate that the ΔG_{max} values of 585-55-FeN_4 (0.45 eV), 484-46-FeN_4 (0.59 eV), 484-48-FeN_4 (0.51 eV), 5775-57-1-FeN_4 (0.60 eV), and 5775-67-FeN_4 (0.45 eV) are higher than that of pristine FeN_4 (0.41 eV), suggesting that these DG-FeN_4 are not conducive to improving NO_3RR performance. Furthermore, the Gibbs free energy diagrams of NO_3RR on the remaining 8 kinds of DG-FeN_4 were depicted in **Figure 3**. One can see that the PDS of 585-58-FeN_4, 484-68-2-FeN_4, 5775-56-FeN_4, and 5775-77-FeN_4 is $^*NO_3$ → $^*HNO_3$ with ΔG_{max} values of 0.29, 0.32, 0.37, and 0.34 eV, respectively, which is agreement with the PDS of pristine FeN_4. Besides, the PDS of 585-56-FeN_4 and 585-68-FeN_4 is *NO → *HNO with ΔG_{max} values of 0.27 and 0.22 eV. On 484-68-1-FeN_4 and 5775-57-2-FeN_4, the PDS is *HNO → *HNOH with ΔG_{max} values of 0.39 and 0.37 eV. Obviously, 585-68-FeN_4 exhibits optimum NO_3RR electrocatalytic activity among 13 kinds of DG-FeN_4. In comparison to pristine FeN_4, the ΔG_{max} value of 585-68-FeN_4 is decreased by 0.19 eV.

2.3. Electrocatalytic Selectivity

During NO_3RR process, H_2 produced from HER is another byproduct that should be suppressed in order to improve the NH_3 Faraday efficiency.[18] We therefore evaluate the competition between NO_3RR and HER on eight DG-FeN_4. In order to suppress HER and achieve a high selectivity, the adsorption Gibbs free energy of $^*NO_3$ should be more negative than that of a H atom. To show the reaction selectivity directly, $\Delta G(^*NO_3)$ versus $\Delta G(^*H)$ and ΔG_{max} of NO_3RR versus $\Delta G(^*H)$ on eight DG-FeN_4 are plotted in **Figure 4**. It is observed that the structures shown in green region in Figure 4a exhibit $\Delta G(^*NO_3) < \Delta G(^*H)$, resulting in the preferential adsorption of $^*NO_3$ on Fe single atom than *H and correspondingly a high selectivity for NO_3RR toward NH_3 synthesis. Conversely, 585-56-FeN_4 in white region in Figure 4a indicates a poor selectivity for NO_3RR, whereas the remaining 7 DG-FeN_4 exhibit a high selectivity for NO_3RR except for 585-56-FeN_4. Considering a stricter criterion, we also make a comparison between ΔG_{max} of NO_3RR and $\Delta G(^*H)$, as illustrated in Figure 4b. Similarly, the structures (585-58-FeN_4 and 585-68-FeN_4) shown in green region in Figure 4b exhibit ΔG_{max} of $NO_3RR < \Delta G(^*H)$, guaranteeing effective inhibition of HER under the limiting potential (U_L) of NO_3RR ($U_L = \Delta G_{max}/e$). To sum up, 585-68-FeN_4 exhibits the most excellent NO_3RR electrocatalytic activity as well as a high selectivity.

2.4. Anti-Dissolution Stability of FeN_4

Demetalation, through dissolution of Fe single atom at the center of FeN_4 active moieties, is considered as one of the most common degradation mechanisms for Fe–N–C catalysts, especially in acidic environments.[19] To assess the anti-dissolution stability of FeN_4 active moieties, we depicted anti-dissolution stability diagrams of pristine FeN_4 and 585-68-FeN_4, as shown in **Figure 5**, in which an equilibrium line can be drawn for Equations (17a)–(17e) (n = 0–4) separating the pH-potential space showing where the FeN_4C structure is more stable than the dissolved Fe^{2+} (gray region in Figure 5). For a given set of temperature (300 K) and Fe^{2+} concentration (10^{-9} M) in Figure 5a, the line of $n = 0$ (horizontal line in Figure 5a, Equation (17a)) represents a purely potential-mediated dissolution reaction. In other words, at potentials below +0.63 V, the FeN_4C structure keeps

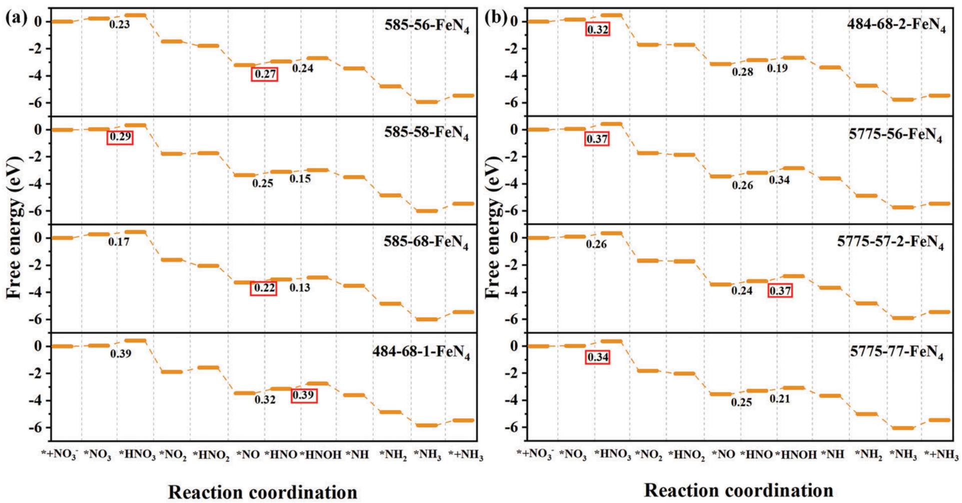

Figure 3. NO_3RR catalytic performance on defective graphene FeN_4. a,b)The Gibbs free energy diagrams for NO_3RR at 300 K when pH = 0, U = 0 V on the surfaces of defective graphene FeN_4.

stable against dissolution to Fe^{2+}, whereas above +0.63 V, the FeN_4C structure is not thermodynamically stable and dissolution to Fe^{2+} will occur. The $n = 2$ line (vertical line in Figure 5a, Equation (17c)) represents a purely pH-mediated dissolution reaction. At pH values smaller than 11.4, dissolution to Fe^{2+} will take place and at pH values greater than 11.4, the FeN_4C structure is stable against dissolution to Fe^{2+}. For the line of $n = 1$ (Equation (17b) where electrons are products), the slope of this line is positive, and below this line, the FeN_4C structure is stable. For the line of $n = 3$ or 4 (Equations (17d) and (17e) where electrons are products), the slopes of these lines are negative, and above these lines, the FeN_4C structure is stable. As a result, an anti-dissolution region (gray region) is constructed where FeN_4C structure keep stable against dissolution to Fe^{2+}. As the concentration of Fe^{2+} increases from 10^{-9} to 10^{-6} M, as illustrated in Figure 5b, the threshold value of potential is increasing to +0.72 V and the threshold value of pH is decreasing to 9.9, leading to an enlarged anti-dissolution region. Likewise, when the concentration of Fe^{2+} increases to 1 M, as shown in Figure 5c, the threshold values of potential and pH change to +0.89 V and 6.9, further rendering an enlarged anti-dissolution region. Comparing pristine FeN_4 and 585-68-FeN_4, it is observed that 585-68-FeN_4 consistently maintains a broader anti-dissolution region than pristine FeN_4 at the same Fe^{2+} concentration. Specifically, when Fe^{2+} concentration equals 10^{-9}, 10^{-6}, and 1 M, the threshold values of potential and pH are +1.18 V and 6.9, +1.27 V and 6.2, +1.45 V and 3.2, respectively. In conclusion, 585-68-FeN_4 presents a more robust anti-dissolution stability than pristine FeN_4. In addition, we also carried out the ab initio molecular dynamics simulations to confirm the dynamic stability of pristine FeN_4 and 585-68-FeN_4 (Figure S9, Supporting Information). The results show that the atoms in these systems vibrate near the initial position and the system energy remains steady throughout the simulation period of 10 ps.

2.5. NO_3RR Activity Origin

In order to further elucidate the origin of high NO_3RR activity on 585-68-FeN_4 and the influence of 5- and 8-member rings on the structure–activity relationship between the electronic structure

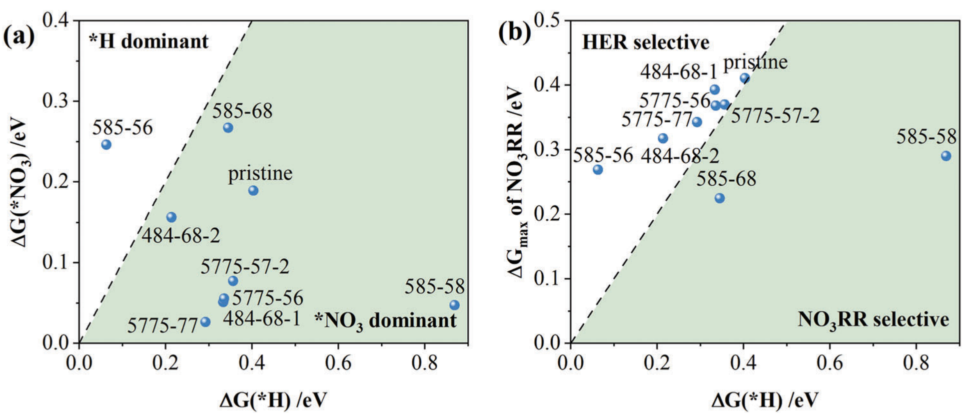

Figure 4. Electrocatalytic selectivity between NO_3RR and HER. a) Adsorption free energy of $*NO_3$ $\Delta G(*NO_3)$ versus adsorption free energy of *H $\Delta G(*H)$. b) The maximum ΔG (ΔG_{max}) of NO_3RR versus $\Delta G(*H)$. The green region represents high selectivity toward NO_3RR.

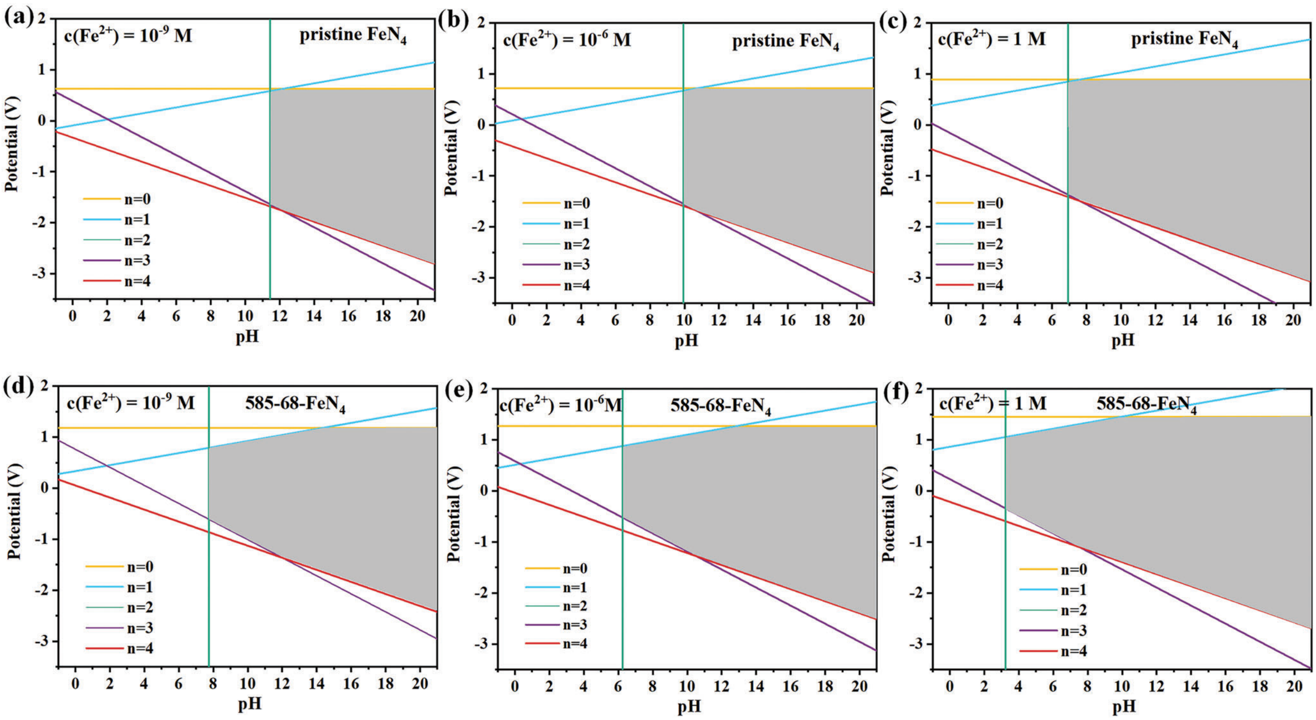

Figure 5. Anti-dissolution stability. Anti-dissolution stability diagrams of a–c) pristine FeN_4 and d–f) 585-68-FeN_4 as a function of pH and potential at 300 K and Fe^{2+} concentrations of 10^{-9}, 10^{-6}, and 1 M. The gray region indicates the anti-dissolution region of FeN_4 against dissolution to Fe^{2+} for all considered reactions (n = 0–4).

and the catalytic activity, the differential charge density and Bader charge analysis calculations were conducted to investigate the electron transfer and redistribution between Fe and N affected by different topological defects. Bader charge calculations show that the valence state of Fe active center in pristine FeN_4, 585-55-FeN_4, 585-56-FeN_4, and 585-68-FeN_4 is +0.952, +0.830, +1.236, and +1.245, respectively. It is noted that, as displayed in **Figure 6**a, the linear relationship for $\Delta G(*NO_3)$ versus the valence state of Fe is found on the abovementioned 4 FeN_4 moieties, where $\Delta G(*NO_3)$ values are gradually increasing (weakened adsorption) as the valence state of Fe increases, which can be expressed in the formula of $\Delta G(*NO_3) = 0.19|\Delta q|+0.02$ with $R^2 = 0.93$. The PDS of 585-55-FeN_4, 585-56-FeN_4, and 585-68-FeN_4 is *NO → *HNO with ΔG_{max} values of 0.45, 0.27, and 0.22 eV, whereas the PDS of pristine FeN_4 is $*NO_3 \rightarrow *HNO_3$ with a ΔG_{max} value of 0.41 eV. It is found in Figure 6b that the ΔG_{max} values are decreasing (enhanced activity) as the valence state of Fe increases, which can be expressed by $\Delta G_{max} = -0.52|\Delta q|+0.89$ with $R^2 = 0.97$. It can therefore be deduced that the high valence state of Fe is conducive to decreasing ΔG_{max} values and further enhancing NO_3RR activity. Correspondingly, the linear relationship between ΔG_{max} and $\Delta G(*NO_3)$ can be expressed as $\Delta G_{max} = -2.64\Delta G(*NO_3)+0.92$ with $R^2 = 0.99$, as shown in Figure 6c. The differential charge density depicted in Figure 6d shows the effect of the topological defects on the charge redistribution of the FeN_4 moieties in pristine FeN_4 and 585-68-FeN_4. A significant charge depletion (blue region) can be visualized above and below the Fe center in 585-68-FeN_4, which is consistent with the higher positive valence state of Fe (+1.245) in 585-68-FeN_4 than that of pristine FeN_4 (+0.952). In addition, in pristine FeN_4, the valence state of N1 (+0.963) is close to that of N3 (+0.992) and N2 (+1.198) is close to N4 (+1.220), exhibiting a high degree of symmetry. However, the valence state of N atoms in 585-68-FeN_4 is +0.808, +1.274, +1.219, and +1.130, indicating that 5- and 8-member rings break the high symmetry of the charge distribution in FeN_4 moiety. As plotted in Figure S10 (Supporting Information), the corresponding cross-sectional views of differential charge density for pristine FeN_4 and 585-68-FeN_4 indicate that the local charge distribution of pristine FeN_4 is more symmetric and the introduction of 585 line defect breaks the symmetric structure of FeN_4 active moiety and induces the distortion of electronic density. In addition, Figure S11 (Supporting Information) plots the PEDs near the Fermi level for pristine FeN_4 and 585-68-FeN_4. It can be seen that 4 N atoms in pristine FeN_4 possess a more symmetric and uniform charge distribution than 585-68-FeN_4.

3. Conclusions

In summary, a theoretical study on 13 kinds of DG-FeN_4 moieties with 585, 484, and 5775 topological line defects for NO_3RR is systematically conducted via density functional theory (DFT) calculations. Regulating FeN_4 active moiety by tailoring the topological defects helps us screen out the topological-defect structures with enhanced NO_3RR activity and find out the structure–activity relationship between electronic structure and catalytic activity. The first hydrogenation step is identified as the potential-determining step of pristine FeN_4 with a ΔG_{max} of 0.41 eV. Among 13 kinds of DG-FeN_4, we found that 8 kinds of DG-FeN_4 are able to reduce

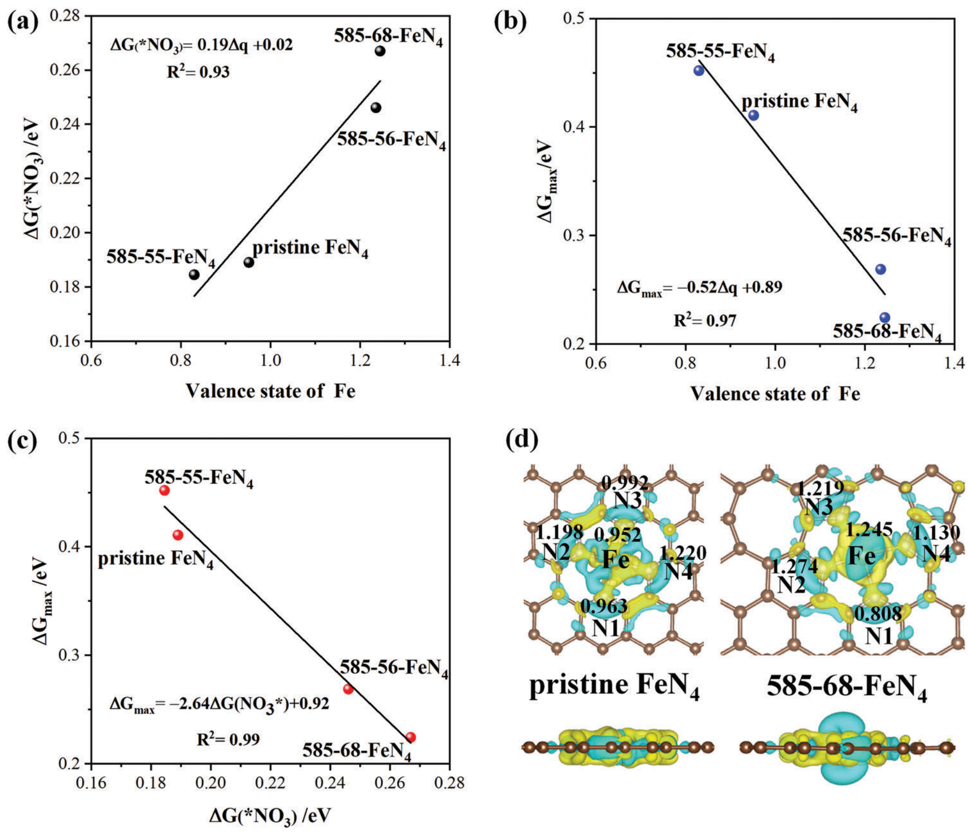

Figure 6. Electronic property analysis. Scaling relationship of a) ΔG(*NO$_3$) versus the valence state of Fe, b) ΔG_{max} versus the valence state of Fe, and c) ΔG_{max} versus ΔG(*NO$_3$) on pristine FeN$_4$, 585-55-FeN$_4$, 585-56-FeN$_4$, and 585-68-FeN$_4$. d) Differential charge density of pristine FeN$_4$ and 585-68-FeN$_4$ with charge accumulation and depletion regions represented in yellow and blue, respectively.

the energy barrier of PDS, exhibiting more excellent NO$_3$RR activity than pristine FeN$_4$. Notably, 585-68-FeN$_4$ presents the optimal NO$_3$RR catalytic performance, with a ΔG_{max} value as low as 0.22 eV. In addition to the high NO$_3$RR activity on 585-68-FeN$_4$, we also demonstrate that 585-68-FeN$_4$ presents a high selectivity against HER by comparing ΔG(*NO$_3$) versus ΔG(*H) and ΔG_{max} of NO$_3$RR versus ΔG(*H). Besides, our anti-dissolution stability diagrams demonstrate a broader anti-dissolution region and more robust anti-dissolution stability on 585-68-FeN$_4$ than pristine FeN$_4$.

The effects of 5- and 8-member rings on FeN$_4$ active moiety are well explained by the variation of the valence state of Fe and charge redistribution on FeN$_4$ moiety. It is proved that the high NO$_3$RR activity on 585-68-FeN$_4$ is ascribed to high valence state of Fe and asymmetric charge distribution on FeN$_4$ moiety. Our work not only provides theoretical guidance for the modulation of graphene topological defects of Fe–N–C SAC for enhanced NO$_3$RR, but also sheds light on the synergistic effects of periodically topological defects and FeN$_4$ active moiety for NO$_3$RR.

4. Experimental Section

Spin-polarized DFT calculations were carried out via the Vienna ab initio simulation package code to provide insights into the NO$_3$RR process on pristine FeN$_4$ and DG-FeN$_4$ moieties.[20] The projector augmented wave method was used to describe the interaction between the ions and the electrons with the frozen-core approximation.[21] The Kohn–Sham valence states were expanded in a plane-wave basis set with a kinetic cutoff energy of 400 eV. Spin-polarized Perdew–Burke–Ernzerhof exchange-correlation functional within generalized gradient approximation calculations were employed to describe the exchange-correlation interaction.[22] For the Brillouin zone integration, a Monkhorst–Pack 3 × 3 × 1 mesh was used for structure optimization, while for electronic structure computations, a 5 × 5 × 1 *k*-points was employed.[23] Dispersion correction method DFT-D3 with Becke–Jonson damping was used to describe the van der Waals interaction.[24] The self-consistent field convergence criteria was set at 1×10^{-4} eV. All structures were fully relaxed until forces on each atom were smaller than 0.05 eV Å^{-1}. To describe the solvent, implicit solvation was considered, in which the dielectric constant was set as 78.4 for water.[25]

A 6 × 6 supercell slab of pristine graphene single layer and three types of defective graphene with 585, 484, and 5775 topological line defects were built as the substrate model. The vacuum gap was set to be 15 Å, which was large enough to avoid interlayer interactions. To anchor Fe single atom, the 2 adjacent carbon atoms were removed and the 4 nearest-neighbored carbon atoms were replaced with N atoms. Eventually, Fe single atom was embedded to construct pristine FeN$_4$ and 13 types of DG-FeN$_4$ moieties.

The binding energies were calculated by the following equation

$$E_b = E_{Fe+substrate} - E_{substrate} - E_{Fe} \quad (1)$$

where $E_{Fe+substrate}$ and $E_{substrate}$ represent the total energies of catalysts with or without Fe single atom, respectively. E_{Fe} represents the energy of an Fe single atom.

The formation energies were calculated by the following equation

$$E_{form} = E_{Fe+substrate} + 6\mu_C - \mu_{Fe} - 4\mu_N - E_{substrate} \quad (2)$$

where $E_{Fe+substrate}$ denotes the total energy of Fe single metal anchored on pristine graphene or defective graphene. μ_{Fe} is the energy of per Fe metal atom in the bulk metal. μ_C and μ_N are the chemical potentials of carbon and nitrogen atom, which were defined as the energy of per atom in pristine graphene and nitrogen gas, respectively. $E_{substrate}$ is the energy of pristine graphene or defective graphene.

Based on the computational hydrogen electrode model, the Gibbs free energy change (ΔG) of NO_3RR on pristine FeN_4 and DG-FeN_4 SACs was evaluated by the formula

$$\Delta G = \Delta E + \Delta ZPE - T\Delta S + \Delta G_U + \Delta G_{pH} + \Delta \int_0^T C_P dT \quad (3)$$

where ΔE is the reaction energy, obtained from DFT calculations, ΔZPE is the zero-point energy change, and $T\Delta S$ is the entropy change upon adsorption. The temperature was set to be 300 K. $\Delta G_U = -neU$, where n is the number of electrons transferred and U is the electrode potential versus the reversible hydrogen electrode. ΔG_{pH} is the correction of the H^+ free energy, which could be evaluated as $\Delta G_{pH} = 2.303 \times k_B T \times pH$. In the DFT studies, the value of pH was assumed to be zero. C_P is the constant-pressure heat capacity, and the integration terms were calculated based on the vibrational energies of NO_3RR intermediates.

NO_3^- electroreduction to NH_3 involved nine protons and eight electrons transferred. The whole reaction could be summarized as

$$NO_3^- + 9H^+ + 8e^- \rightarrow NH_3 + 3H_2O \quad (4)$$

Involving the intermediates $*NO_3$, $*HNO_3$, $*NO_2$, $*HNO_2$, $*NO$, $*NOH/*HNO$, $*N/*HNOH/*H_2NO$, $*NH/*H_2NOH$, $*NH_2$, and $*NH_3$, the elementary steps of NO_3^- electroreduction to NH_3 were described as below

$$* + NO_3^- \rightarrow *NO_3 + e^- \quad (5)$$

$$*NO_3 + H^+ + e^- \rightarrow *HNO_3 \quad (6)$$

$$*HNO_3 + H^+ + e^- \rightarrow *NO_2 + H_2O \quad (7)$$

$$*NO_2 + H^+ + e^- \rightarrow *HNO_2 \quad (8)$$

$$*HNO_2 + H^+ + e^- \rightarrow *NO + H_2O \quad (9)$$

$$*NO + H^+ + e^- \rightarrow *HNO \quad (10a)$$

$$*NO + H^+ + e^- \rightarrow *NOH \quad (10b)$$

$$*HNO + H^+ + e^- \rightarrow *HNOH \quad (11a)$$

$$*HNO + H^+ + e^- \rightarrow *H_2NO \quad (11b)$$

$$*NOH + H^+ + e^- \rightarrow *N + H_2O \quad (11c)$$

$$*NOH + H^+ + e^- \rightarrow *HNOH \quad (11d)$$

$$*HNOH + H^+ + e^- \rightarrow *NH + H_2O \quad (12a)$$

$$*HNOH + H^+ + e^- \rightarrow *H_2NOH \quad (12b)$$

$$*H_2NO + H^+ + e^- \rightarrow *H_2NOH \quad (12c)$$

$$*N + H^+ + e^- \rightarrow *NH \quad (12d)$$

$$*NH + H^+ + e^- \rightarrow *NH_2 \quad (13a)$$

$$*H_2NOH + H^+ + e^- \rightarrow *NH_2 + H_2O \quad (13b)$$

$$*NH_2 + H^+ + e^- \rightarrow *NH_3 \quad (14)$$

where * denotes the active site on catalyst.

To avoid calculating the energy of charged NO_3^- directly, gaseous $HNO_3(g)$ was chosen as a reference. Therefore, $\Delta G(*NO_3)$ was obtained by

$$\Delta G(*NO_3) = G(*NO_3) - G(*) - [G(HNO_3) - \frac{1}{2}G(H_2)] + \Delta G_{correct} \quad (15)$$

where $\Delta G(*NO_3)$ is the total Gibbs free energy of NO_3 adsorbed on SAC, and $G(*)$ is the Gibbs free energy of SAC. $G(HNO_3)$ and $G(H_2)$ denote the Gibbs free energy of gaseous HNO_3 and H_2 molecules, respectively. The free energy correction $\Delta G_{correct}$ was set as 0.392 eV.[7b,26]

Acid anti-dissolution stability diagrams of FeN_4C were plotted via the method proposed by Holby et al.[27] For FeN_4C structures considered here, a generalized reaction for dissolution of Fe single atom could be written as follows

$$FeN_4C + nH^+_{aq} \rightarrow Fe^{2+}_{aq} + N_4CH_n + (2 - n)e^- \quad (16)$$

When n varied from 0 to 4, it was available to determine a set of dissolution reactions that would define Fe anti-dissolution stability as a function of pH and potential following Equation (17). The corresponding dissolution reactions were outlined in Equations (17a)–(17e)

$$FeN_4C \rightarrow Fe^{2+}_{aq} + N_4C + 2e^- \quad (17a)$$

$$FeN_4C + H^+_{aq} \rightarrow Fe^{2+}_{aq} + N_4CH_1 + e^- \quad (17b)$$

$$FeN_4C + 2H^+_{aq} \rightarrow Fe^{2+}_{aq} + N_4CH_2 \quad (17c)$$

$$FeN_4C + 3H^+_{aq} + e^- \rightarrow Fe^{2+}_{aq} + N_4CH_3 \quad (17d)$$

$$FeN_4C + 4H^+_{aq} + 2e^- \rightarrow Fe^{2+}_{aq} + N_4CH_4 \quad (17e)$$

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords

density functional theory, electrocatalysts, iron–nitrogen–carbon, nitrate reduction reaction, periodic topological defects

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